Practice Problems 6: Solutions

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

Exercise 5.2

1. (c)

$$
\begin{aligned}
&8\begin{vmatrix} 0 & 1 \\ 0 & 3 \end{vmatrix} - 1\begin{vmatrix} 4 & 1 \\ 6 & 3 \end{vmatrix} + 3\begin{vmatrix} 4 & 0 \\ 6 & 0 \end{vmatrix} \\
=&8(0-0) - 1(12-6) + 3(0-0) \\
=&0 - 6 + 0 \\
=&-6
\end{aligned}
$$

(e)

$$
\begin{aligned}
&a\begin{vmatrix} c & a \\ a & b \end{vmatrix} - b\begin{vmatrix} b & a \\ c & b \end{vmatrix} + c\begin{vmatrix} b & c \\ c & a \end{vmatrix} \\
=&a\left(cb - a^2\right) - b\left(b^2 - ac\right) + c\left(ab - c^2\right) \\
=&abc - a^3 - b^3 + abc + abc - c^3 \\
=&-a^3 - b^3 - c^3 + 3abc
\end{aligned}
$$

(f)

$$
\begin{aligned}
&x\begin{vmatrix} y & 2 \\ -1 & 8 \end{vmatrix} - 5\begin{vmatrix} 3 & 2 \\ 9 & 8 \end{vmatrix} + 0\begin{vmatrix} 3 & y \\ 9 & -1 \end{vmatrix} \\
=&x(8y+2) - 5(24-18) + 0(-3-9y) \\
=&8xy + 2x - 30 + 0 \\
=&8xy + 2x - 30
\end{aligned}
$$

2. $|C_{13}| : 1+3=4$ is even so $+$
   $|C_{23}| : 2+3=5$ is odd so $-$
   $|C_{33}| : 3+3=6$ is even so $+$
   $|C_{41}| : 4+1=5$ is odd so $-$
   $|C_{34}| : 3+4=7$ is odd so $-$

3. Minor of a:

$$
|M_{11}| = \begin{vmatrix} e & f \\ h & i \end{vmatrix} = ei - fh
$$

Cofactor of a:

$$|C_{11}| = (-1)^{1+1}\,|M_{11}| = |M_{11}|$$

Minor of b:

$$|M_{12}| = \begin{vmatrix} d & f \\ g & i \end{vmatrix} = di - fg$$

Cofactor of b:

$$|C_{12}| = (-1)^{3}\,|M_{12}| = -\,|M_{12}|$$

Minor of f:

$$|M_{23}| = \begin{vmatrix} a & b \\ g & h \end{vmatrix} = ah - bg$$

Cofactor of f:

$$|C_{23}| = (-1)^{5}\,|M_{23}| = -\,|M_{23}|$$

6. Minors of third row:

$$|M_{31}| = \begin{vmatrix} 11 & 4 \\ 2 & 7 \end{vmatrix} = 69, \quad |M_{32}| = \begin{vmatrix} 9 & 4 \\ 3 & 7 \end{vmatrix} = 51, \quad |M_{33}| = \begin{vmatrix} 9 & 11 \\ 3 & 2 \end{vmatrix} = -15$$

Cofactors: $|C_{31}| = |M_{31}|, |C_{32}| = -|M_{32}|, |C_{33}| = |M_{33}|$.